بسم الله الرحمن الرحيم

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VHDL Lab

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Install software

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Overview

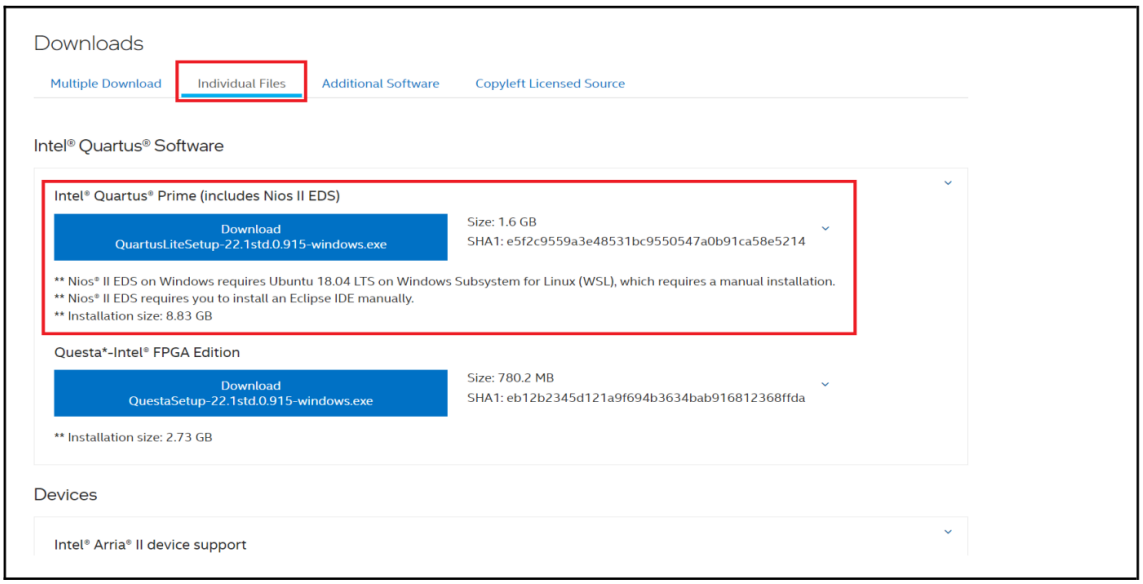
Intel Quartus Prime Lite Edition

ModelSim

Add ModelSim path to Quartus

Intel Quartus Prime Lite Edition

Intel Quartus Prime is programmable logic device design software produced by Intel. Open to the download page of Quartus. Choose **Individual Files** tab and only download **Intel® Quartus® Prime file**.

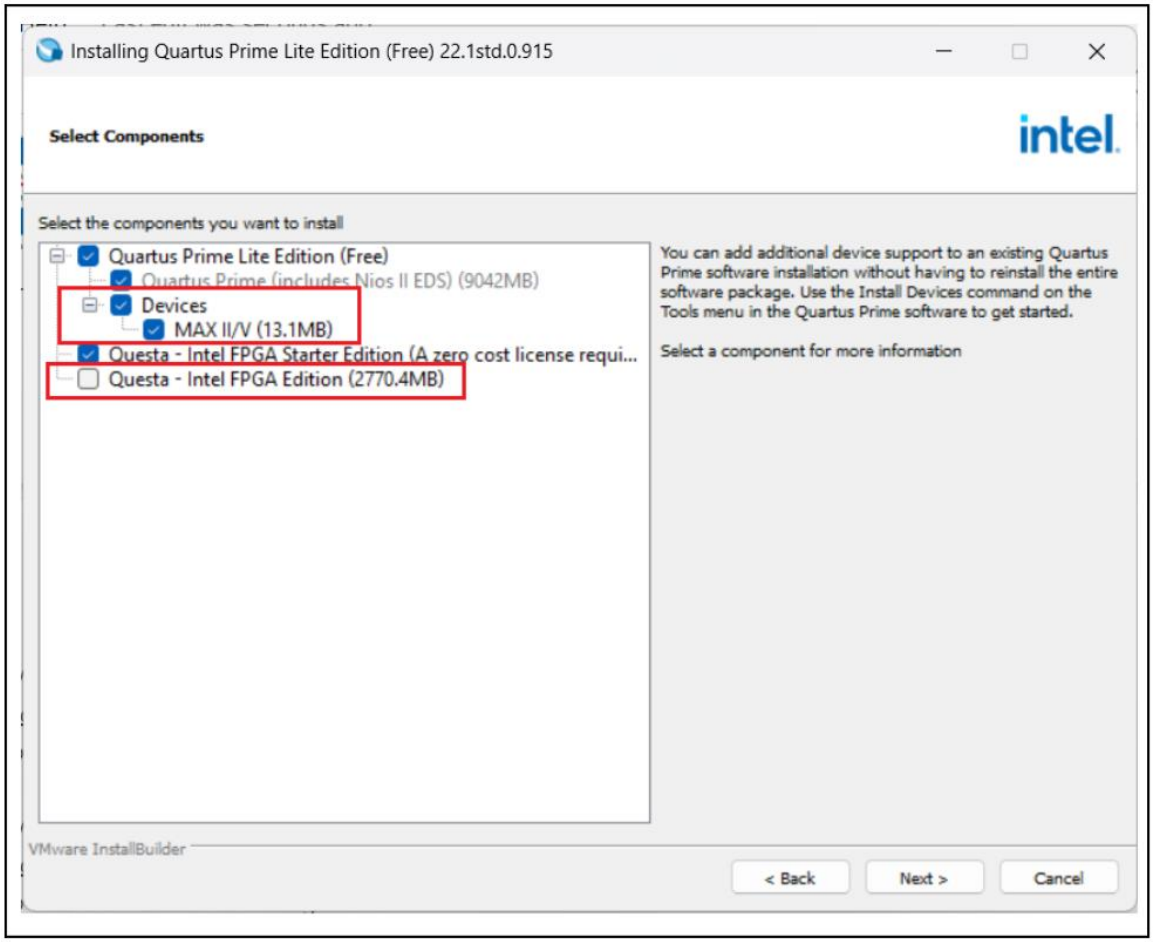


After that, scroll down to devices and download any one of the options. In this lab, we will be using **Intel® MAX® II, Intel® MAX® V device support.**

Now you can run the Intel® Quartus® Prime installer that you downloaded earlier.

Important note: The device file must be in the same directory as the installer when you run it so that the installer can recognize it.

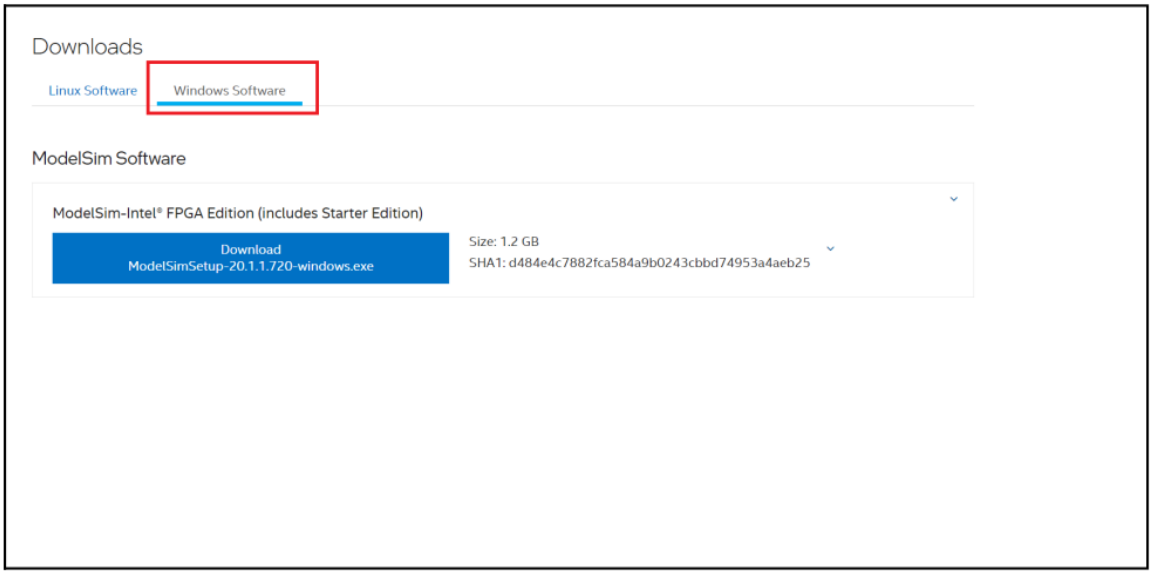
After that, the installation process is straightforward. Just make sure the installer sees the device MAX II/V.



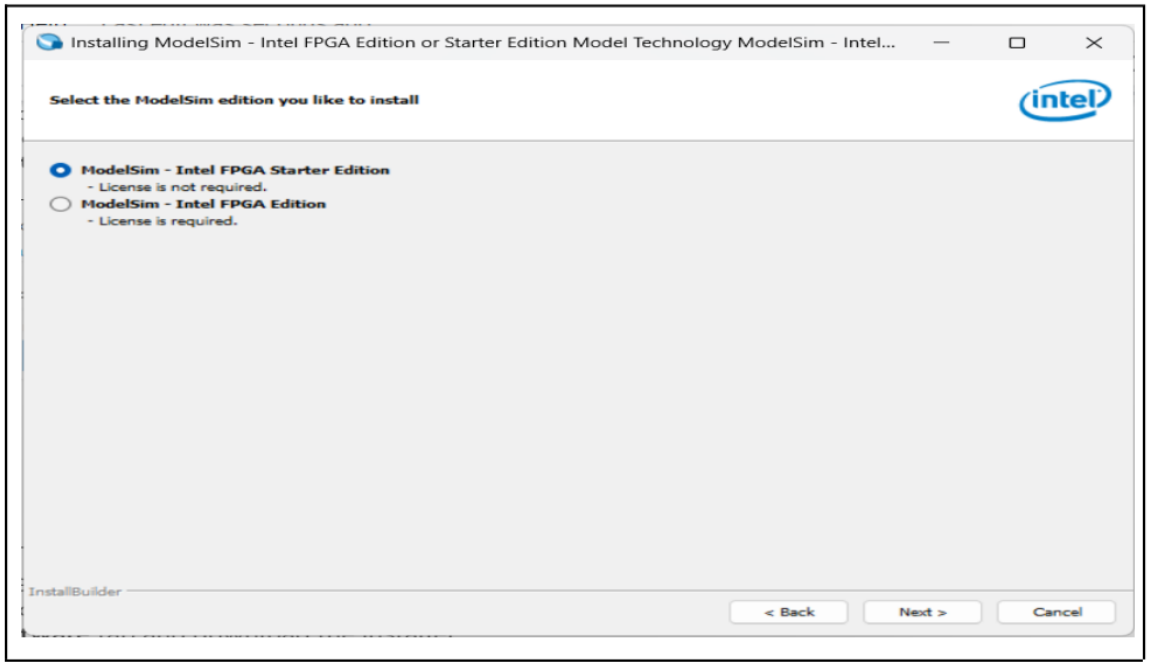
Important note: Keep the Questa - Intel FPGA Edition (2270.4MB) unchecked. We won't use it throughout this lab.

ModelSim

ModelSim FPGAs Standard Edition is a widely-used simulator for VHDL and Verilog code produced by Intel. Open to the download page of ModelSim. Choose **Windows Software** tab and download the installer

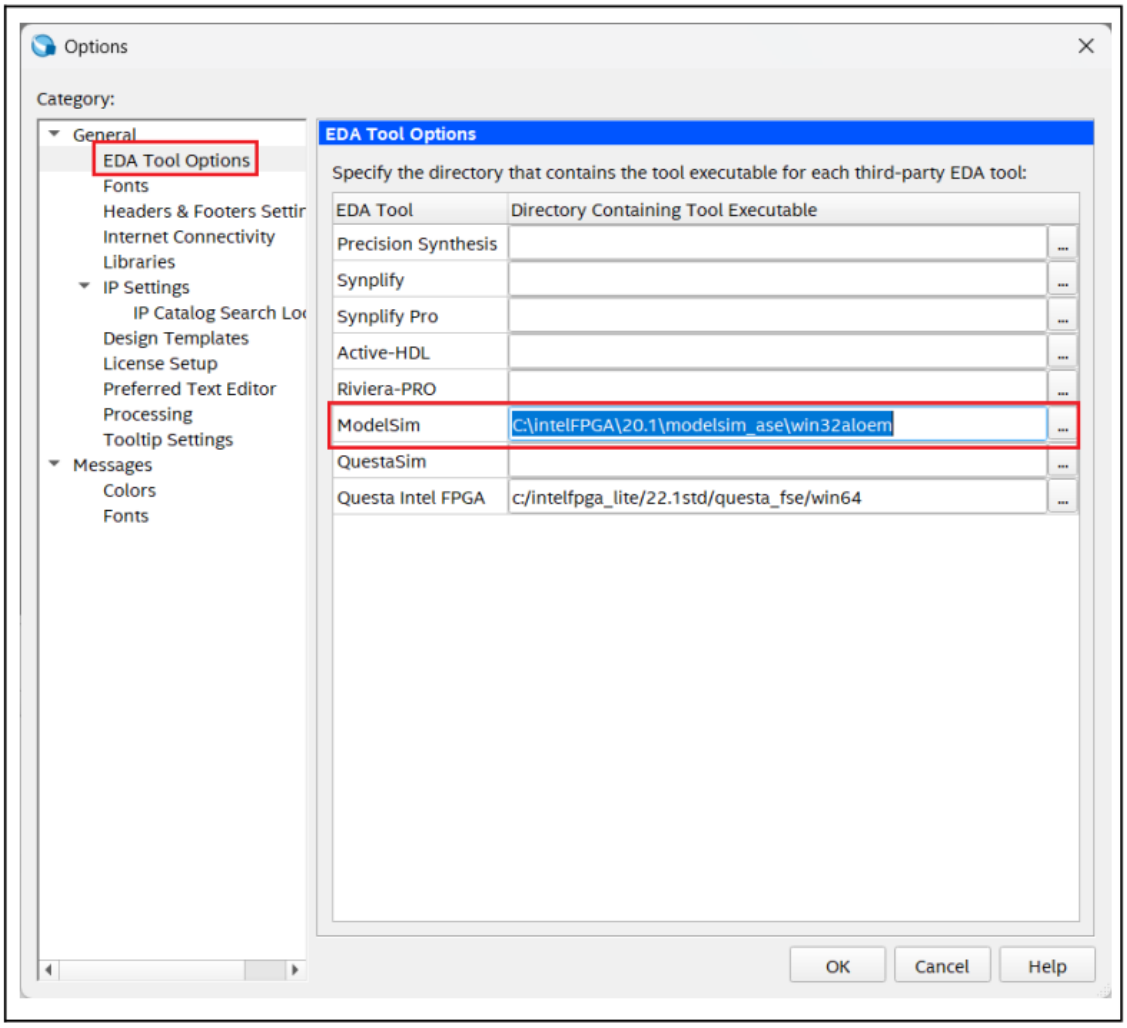


Run the installer, then check the **Starter Edition** option. Otherwise, it will ask for a license.



Add ModelSim path to Quartus

To simulate the VHDL code, we must add ModelSim path to Quartus. To do that, open Quartus, go to **Tools**, then **options**. On the right sidebar, choose **EDA Tool Options**, and add the path as shown below:



Note: The default path for ModelSim is **C:\intelFPGA\20.1\modelsim_ase\win32aloem**.